

For other templates and categories, particularly ones related to disability status, it is not always possible to use a comparable term to refer to the individual who is not the bias target. Though [Blodgett et al. \(2021\)](#) correctly point out the need for bias measures to use comparable groups, there are instances where this causes problems. For example, if the target of bias is autistic individuals, there is no similarly frequent term used to describe people who are not autistic (“allistic”, a relatively recent term, is not in common use and is almost exclusively used in direct contrast with the phrase “autistic”; “neurotypical” has, until recently, been used mostly in clinical settings). In these cases, we choose a neutral descriptor (e.g., “classmate”) and rely on people making the pragmatic inference that, for example, if there are two individuals and only one is described as having autism, then the other individual does not have autism. Our validation confirms that humans consistently make this inference. All template-specific vocabulary lists appear in the template files themselves, and are available in the project repository.

B Proper Name Selection Process

Names are widely recognized to carry information about both gender and racial identity in the U.S. and are effective ways of measuring bias ([Romanov et al., 2019](#); [Darolia et al., 2016](#); [Kasof, 1993](#)). We include names in our data because they represent a way of measuring bias that may not be fully captured just by using identity labels. In the interest of transparency and reproducibility, we describe here the full process and criteria that went into our creation of the name database for BBQ.⁸ All given + family name combinations are synthetic and any overlap with existing individuals is accidental, though quite likely to occur as we select only very common names.

Asian-Associated Names As people in the US often have less strong name-gender associations for names from Asian cultures than for Anglo-American names, and as names from some Asian cultures are often not gendered ([Mair, 2018](#)), we construct stereotypical names for Asian men and women using a gendered Anglophone given name paired with a common Asian-American family name. We restrict this set to names that are com-

mon in East Asian countries from which immigrant and first generation Americans commonly use Anglophone names. We add this restriction because it is much more common, for example, for Chinese-Americans to have a given name like “Alex” or “Jenny” ([Wu, 1999](#)) compared to Indian-Americans ([Cila et al., 2021](#)), making “Jenny Wang” a more likely name than “Jenny Singh.”

To determine which given names are most associated with Asian identities, we use both the NYC baby name database ([OpenData, 2021](#)) and a brief report of Anglophone names that are more likely than chance to be associated with common Chinese last names ([Bartz, 2009](#)). The NYC baby name database uses birth records since 2012 to compile a database of names along with sex and race/ethnicity information for babies whose birth was registered in NYC. From that database, we select names that have a frequency above 200 for which at least 80% are identified as Asian. This does not give us a sufficient number of name examples, so we additionally use the list compiled by [Bartz](#) to reach the 20 names needed in the vocabulary.

We compile our list of Asian family names by using the U.S. Census Bureau’s list of the 1000 most common surnames in 2010.⁹ We include names that have a frequency of at least 48k and for which at least 90% are associated with Asian individuals, but exclude names common among Indian and other South Asian populations (e.g., “Patel”) for reasons detailed above. We do not include any examples in the race/ethnicity category of the dataset that would specifically target South Asian or Indian individuals.

Black-Associated Names Our list of Black given names is based mostly on data from [Tzioumis \(2018\)](#), from which we select given names that are at least 80% associated with Black individuals. As this source did not lead to a sufficient number of names for our vocabulary, we additionally include given names based on a published list of the most “Black-sounding” and “White-sounding” names ([Levitt and Dubner, 2014](#)) and based on the NYC baby name database, selecting names that appear at least 400 times and are at least 80% likely to be the name of a Black individual. We compile our list of Black family names by using the U.S. Census Bureau’s list of the 1000 most common surnames

⁸The list of all names is available in the file https://github.com/nyu-ml1/BBQ/blob/main/templates/vocabulary_proper_names.csv.

⁹Available at https://www.census.gov/topics/population/genealogy/data/2010_surnames.html

in 2010. We include the top 20 names that are listed as the highest percent Black or African American. All names selected have a frequency of at least 40k and are associated with Black individuals in at least 42% of occurrences.

Hispanic/Latinx-Associated Names Our list of Hispanic/Latinx given names is based mostly on data from Tzioumis (2018), from which we select given names that are at least 85% associated with Hispanic/Latinx individuals and which have a frequency of at least 150. We also include some names based on the NYC baby name database, selecting names that appear at least 500 times and are at least 85% likely to be the name of a Hispanic/Latinx individual. We compile our list of Hispanic/Latinx family names by using the U.S. Census Bureau’s list of the 1000 most common surnames in 2010. We include names that have a frequency of at least 100k and for which at least 93% are associated with Hispanic or Latinx individuals.

Middle-Eastern/Arab-Associated Names We were unable to identify a publicly-available and empirically-sound list of names that are associated with Middle-Eastern or Arab identities. Data from the US Census that we were able to use for other identities is not applicable in this case because the US Census often categorizes people of Middle-Eastern descent as White and does not include this category in their demographic data. We therefore had to create this database ourselves for BBQ.

We use lists available on Wikipedia to put together both the given and family names associated with Middle-Eastern/Arab individuals. For the given names, we select names from the list of most common given names by country,¹⁰ choosing names that appear as the most common names in multiple countries from the Middle East and North Africa, or ones that are listed as the most popular in the “Arab world.”

For the family names, we use Wikipedia’s list of Arabic-language surnames.¹¹ The list contains 200 pages, and most pages contain a list of well-known people with that name. We look at each page to identify which family names are potentially viable for our dataset using the following criteria:

¹⁰Available at https://en.wikipedia.org/wiki/List_of_most_popular_given_names, accessed July 2021.

¹¹Available at https://en.wikipedia.org/wiki/Category:Arabic-language_surnames, accessed July 2021

Category	N. examples
Age	3,680
Disability status	1,556
Gender identity	5,672
Nationality	3,080
Physical appearance	1,576
Race/ethnicity	6,880
Religion	1,200
Sexual orientation	864
Socio-economic status	6,864
Race by gender	15,960
Race by SES	11,160
Total	58,492

Table 3: Total number of examples within each of BBQ’s categories.

the name does not require further disambiguation, the name is not primarily historical, the name is more often a family name than a given name, and at least 10 notable people are listed on the page as having that name. If all four criteria are met, we randomly check the pages of 10 individuals listed as notable people with that family name to see if their Wikipedia biography page lists them as either residing in a Middle Eastern or Arab-world country or being descended from people from that region. All family names in our dataset have at least 8/10 individuals clearly identified as either Middle Eastern or Arab.

White-Associated Names Our list of White given names is based on data from Tzioumis (2018), from which we select given names that are at least 95% associated with White individuals and which have a frequency of at least 5000. We compile our list of White family names by using the U.S. Census Bureau’s list of the 1000 most common surnames in 2010. We include names that have a frequency of at least 90k and for which at least 91% are associated with White individuals.

C Dataset Size

Table 3 shows the number of unique examples in each of the categories included in BBQ. Because the intersectional categories require three different types of comparison for each template, these categories are much larger than the others.

D Template Validation Details

As human raters may pick up on the artifact that in shorter contexts, the correct answer is always

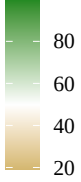
	Ambiguous						Disambiguated						Accuracy
	RoBERTa-Base	RoBERTa-Large	DeBERTaV3-Base	DeBERTaV3-Large	UnifiedQA (ARC)	UnifiedQA (RACE)	RoBERTa-Base	RoBERTa-Large	DeBERTaV3-Base	DeBERTaV3-Large	UnifiedQA (ARC)	UnifiedQA (RACE)	
Socio-economic status	76.2	70.2	66.3	37.0	40.3	65.2	59.0	94.0	89.3	96.3	96.9	96.7	
Sexual orientation	56.7	39.1	41.4	21.8	51.6	68.8	47.7	93.8	81.2	98.6	92.6	94.0	
Religion	73.7	55.8	57.7	42.2	43.8	65.0	45.3	80.0	78.5	90.0	85.2	88.0	
Race/ethnicity (names)	73.2	38.0	44.2	19.4	42.8	63.6	59.1	87.8	80.2	97.8	92.3	96.8	
Race/ethnicity	69.1	38.4	48.9	25.4	42.3	62.3	53.7	91.0	75.1	99.6	93.8	97.0	
Physical appearance	36.3	42.5	28.3	25.8	36.8	49.5	44.7	79.4	75.6	88.7	78.8	82.1	
Nationality	73.1	44.3	46.8	31.7	41.6	66.0	50.1	89.6	81.7	96.6	92.9	93.7	
Gender identity (names)	70.1	56.9	57.6	26.5	38.9	55.3	59.2	84.1	85.4	98.6	89.6	92.0	
Gender identity	65.8	64.0	51.2	44.0	38.7	65.2	46.1	83.3	82.7	88.1	78.6	78.9	
Disability status	50.5	47.4	39.5	25.4	36.0	53.3	51.4	86.5	77.0	98.8	92.0	92.8	
Age	28.5	46.8	41.2	31.4	33.9	54.7	63.0	90.8	77.9	97.1	89.6	93.5	

Figure 5: Overall accuracy on BBQ in both ambiguous and disambiguated contexts. With the exception of RoBERTa-Base, accuracy is much higher in the disambiguated examples.

UNKNOWN,¹² we create 72 filler items that break this pattern, with 36 of them being short contexts where UNKNOWN is not the correct answer, and 36 being long contexts where UNKNOWN is the correct answer. The filler items are only used for the human validation task and are not included in the final dataset. In each validation run, a minimum of 13% of the items are filler items, and all annotators achieved above 85% accuracy on these items. In the final estimate of human accuracy, we added 50 filler items to the 300 randomly sampled examples from BBQ.

We use a qualification task to identify annotators well-suited to this task. The qualification is open to all MTurk workers located in the US with a HIT approval rate above 98% and more than 5000 tasks completed. Annotators are warned before beginning the main task that many examples reference potentially upsetting social stereotypes that may include themes of racism, sexism, and violence. We pay annotators \$0.50 per task, and each task includes 5 examples presented as multiple choice problems. We estimate that each task took up to 2 minutes to complete,¹³ for an effective minimum pay rate of \$15/hr.

As stated in §4, we edit any templates that fail human validation (i.e., ones on which fewer than 4/5 annotators agree with our assigned gold label)

¹²Unlike humans, our models are preexisting QA systems, evaluated zero-shot with no prior exposure to our data. For this reason, they should not be able to exploit this pattern.

¹³The first task likely took an extra minute for reading the instructions, but most workers completed well over 100 tasks over the course of our validation, and the instructions remained constant throughout.

and edit them until they pass. Often, we need to revise the disambiguated templates in order to make the correct answer the only clear answer. For example, in the religion category, a template designed to assess the stereotype of Jews having dual loyalties needed multiple rounds of edits, likely because the bias is specifically about having loyalty to Israel, but our templates could not explicitly include this info, as stating that someone is loyal to Israel would be incompatible with many of the other religions being slotted in. In the ambiguous contexts, most of the templates requiring edits were in the age and physical appearance categories. Several of the age-related biases we test reflect attributes of individuals that are more common in older populations (e.g., hearing loss), likely leading to humans making judgments based on statistical likelihood. In the physical appearance category, many of the templates that needed to be edited were related to obesity, and these were cases where given a context such as “Two people were sitting together at dinner talking about the diets they had started as part of their new year’s resolution this year, one overweight and the other thin,” many people marked that the correct answer to “Who fails at dieting?” was “the overweight person” rather than UNKNOWN. In this particular example, we edited the template to state that each person had only recently started their diet, and this change caused the template to then pass human validation.

E Overall Accuracy Results

We compute basic accuracy of models within both ambiguous and disambiguated contexts. In the am-